



# **NATIONAL CIVIL REGISTRY ENTERPRISE NETWORK ZAMBIA**

**UNDER THE SUPERVISION OF ENG. ABDELRAHMAN SOLIMAN**

# PROJECT TEAM



Mohamed Abdelhamid :

- Team Leader
- Project Plan
- Vlan\_IP structure
- OSPF
- ABRs
- VoIP
- HSRP In Servers
- FTP
- DHCP
- Syslog
- HTTP
- DHCP spoofing

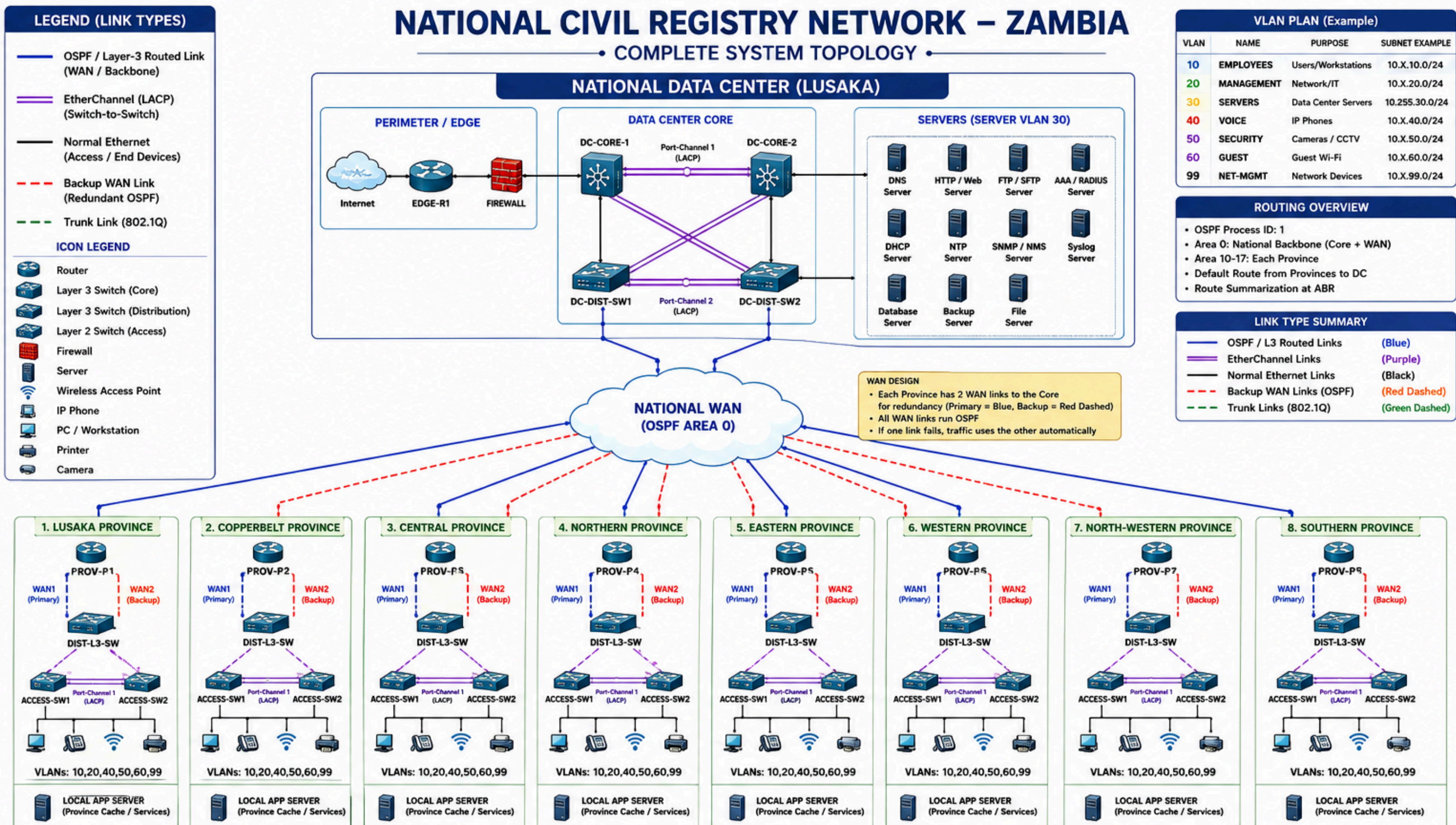
Ahmed Samir :

- Truncking Switches
- Access Switching
- ACL
- interVlanRouting
- WLAN
- SNMP
- Portfast BPDU Guard
- VPN

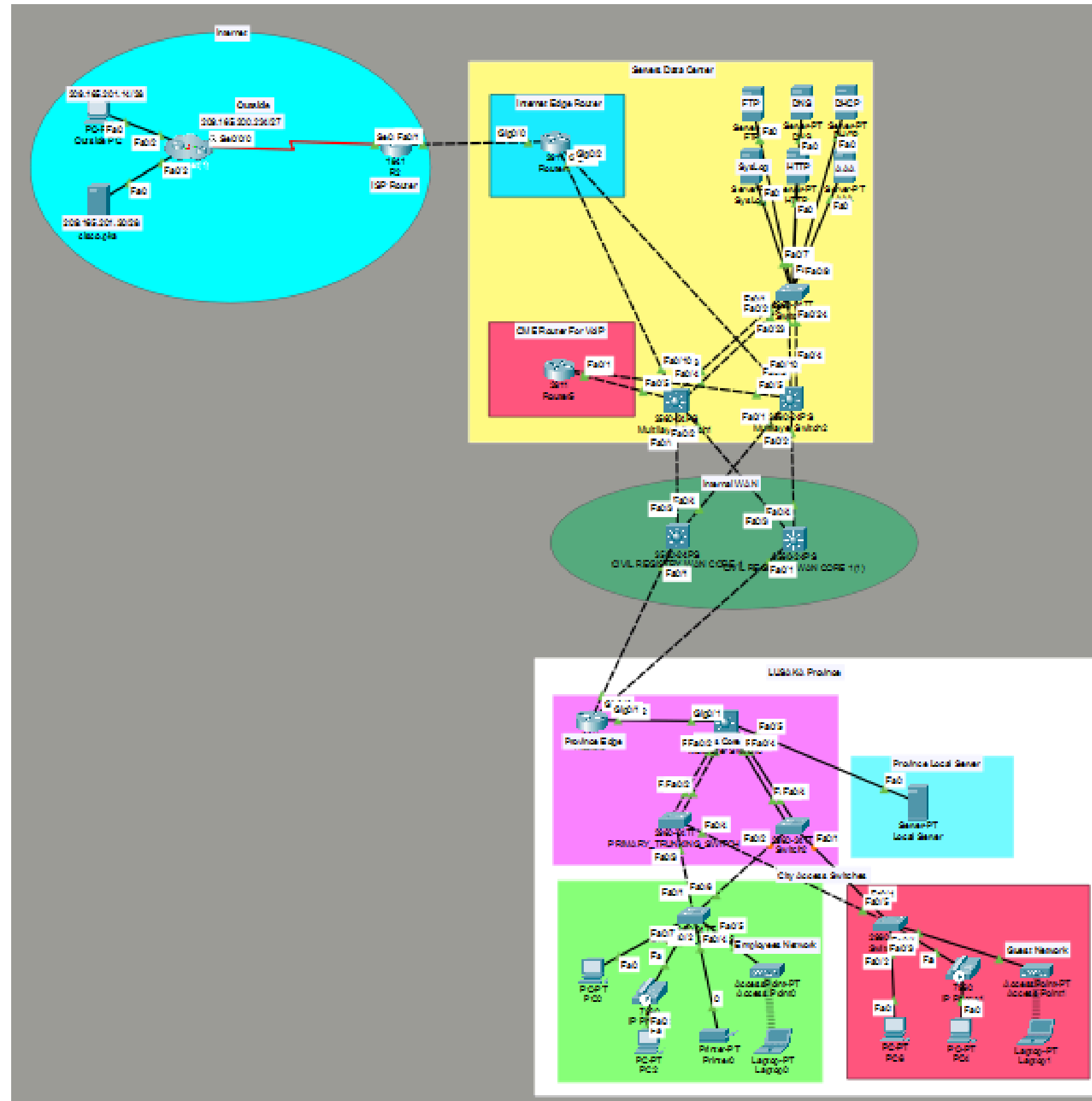
Mohamed Nael :

- Servers
- Eth Channel
- Nat
- Firewall
- DNS
- AAA
- FTP
- DAI

# PROJECT ARCHITECTURE WITH AI



# PACKET TRACER IMPLEMENTATION



# PROJECT VLANS ARCHITECTURE



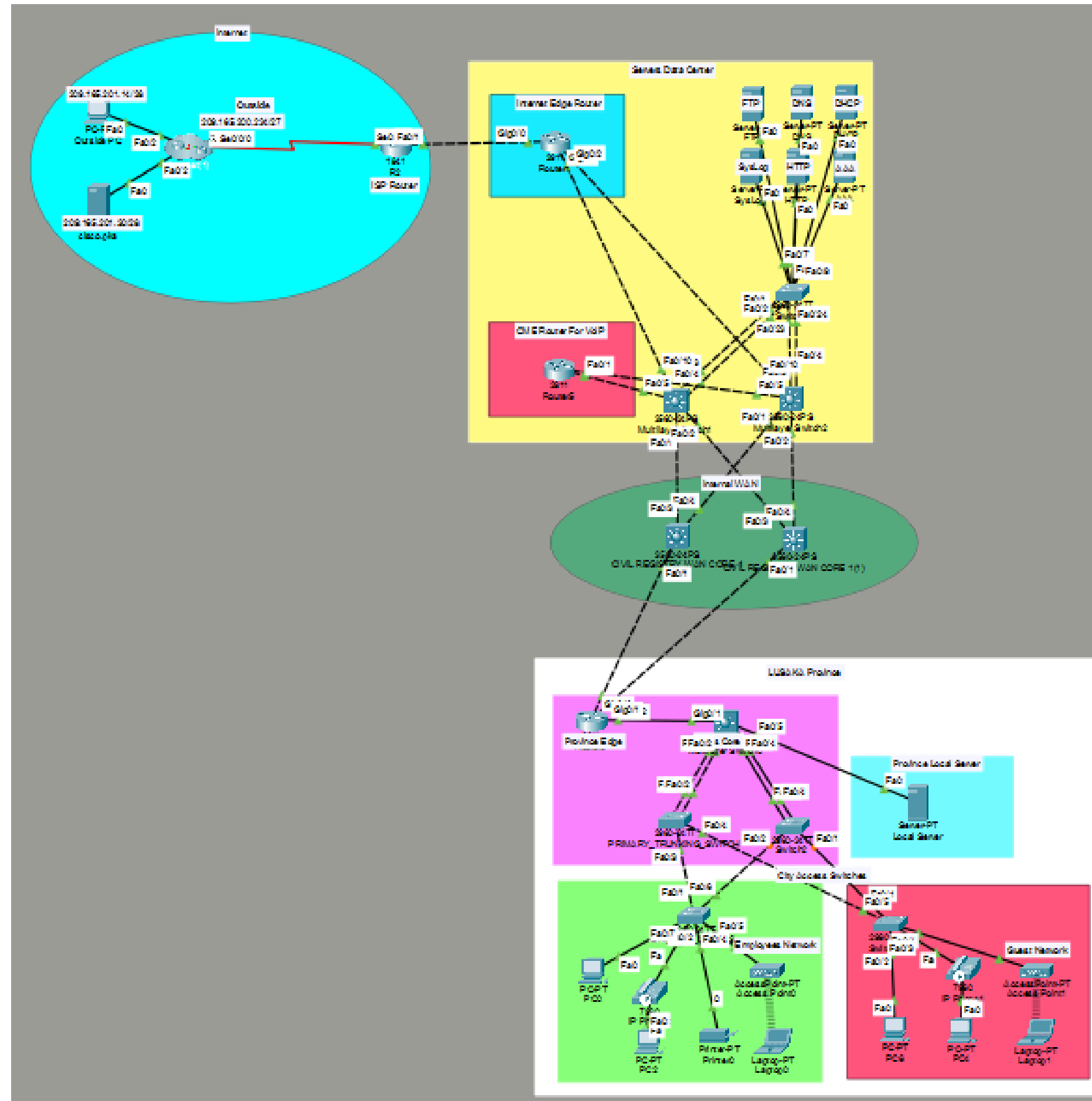
| VIAN | Name           | IP Per Province | Cacacity Per Province |
|------|----------------|-----------------|-----------------------|
| 10   | Employee       | 10.x.8.0 /21    | 2046                  |
| 20   | Guest          | 10.x.16.0 /22   | 1022                  |
| 30   | Servers        | 10.x.30.0 /24   | 255                   |
| 40   | Voice          | 10.x.40.0 /24   | 255                   |
| 50   | Reserved       | 10.x.50.0 /24   | 255                   |
| 60   | Management     | 10.x.60.0 /24   | 255                   |
| 99   | Net Management | 10.x.99.0 /24   | 255                   |

# PROVENCE IP ARCETECTURE AND AREA

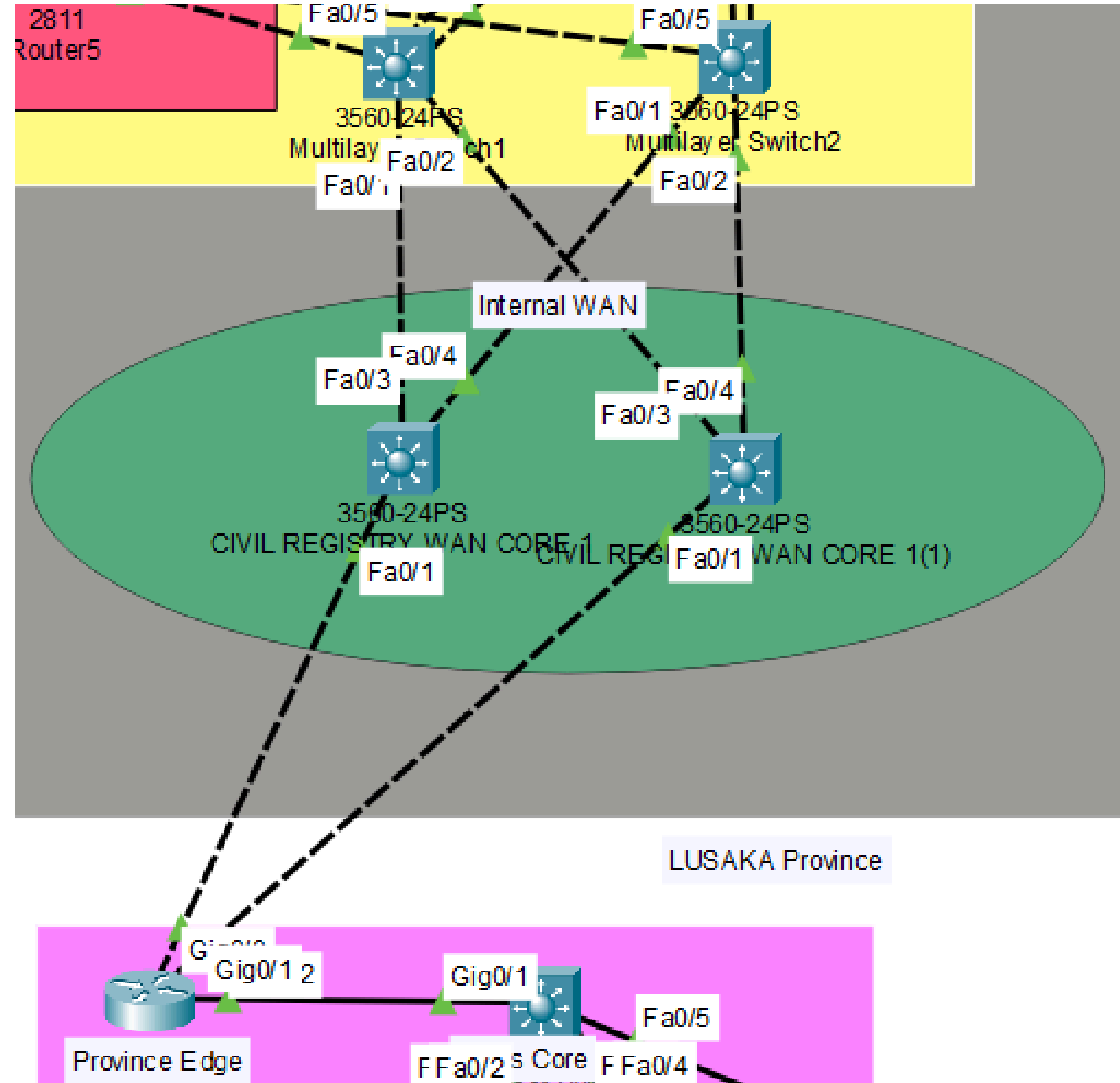


| Area | Name           | IP Per Province | USABLE IPs Per Province |
|------|----------------|-----------------|-------------------------|
| 0    | WAN & DC       | 10.255.0.0 /16  | 65634                   |
| 10   | Prov1 : LUSAKA | 10.1.0.0 /16    |                         |
| 20   | Prov2          | 10.2.0.0 /16    |                         |
| 30   | Prov3          | 10.3.0.0 /16    |                         |
| 40   | Prov4          | 10.4.0.0 /16    |                         |
| 50   | Prov5          | 10.5.0.0 /16    |                         |
| 60   | Prov6          | 10.6.0.0 /16    |                         |
| 70   | Prov7          | 10.7.0.0 /16    |                         |
| 80   | Prov8          | 10.8.0.0 /16    |                         |

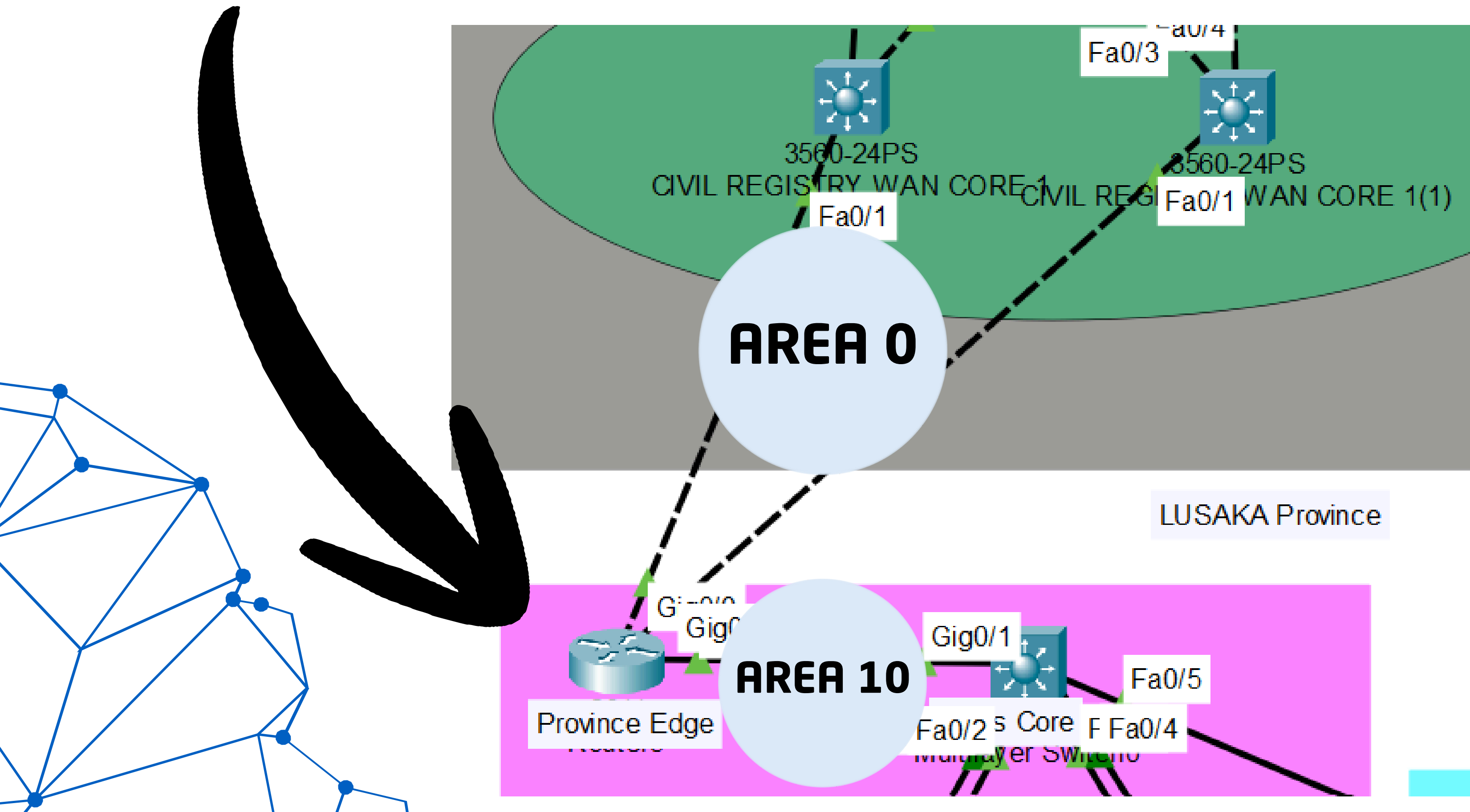
# PACKET TRACER IMPLEMENTATION



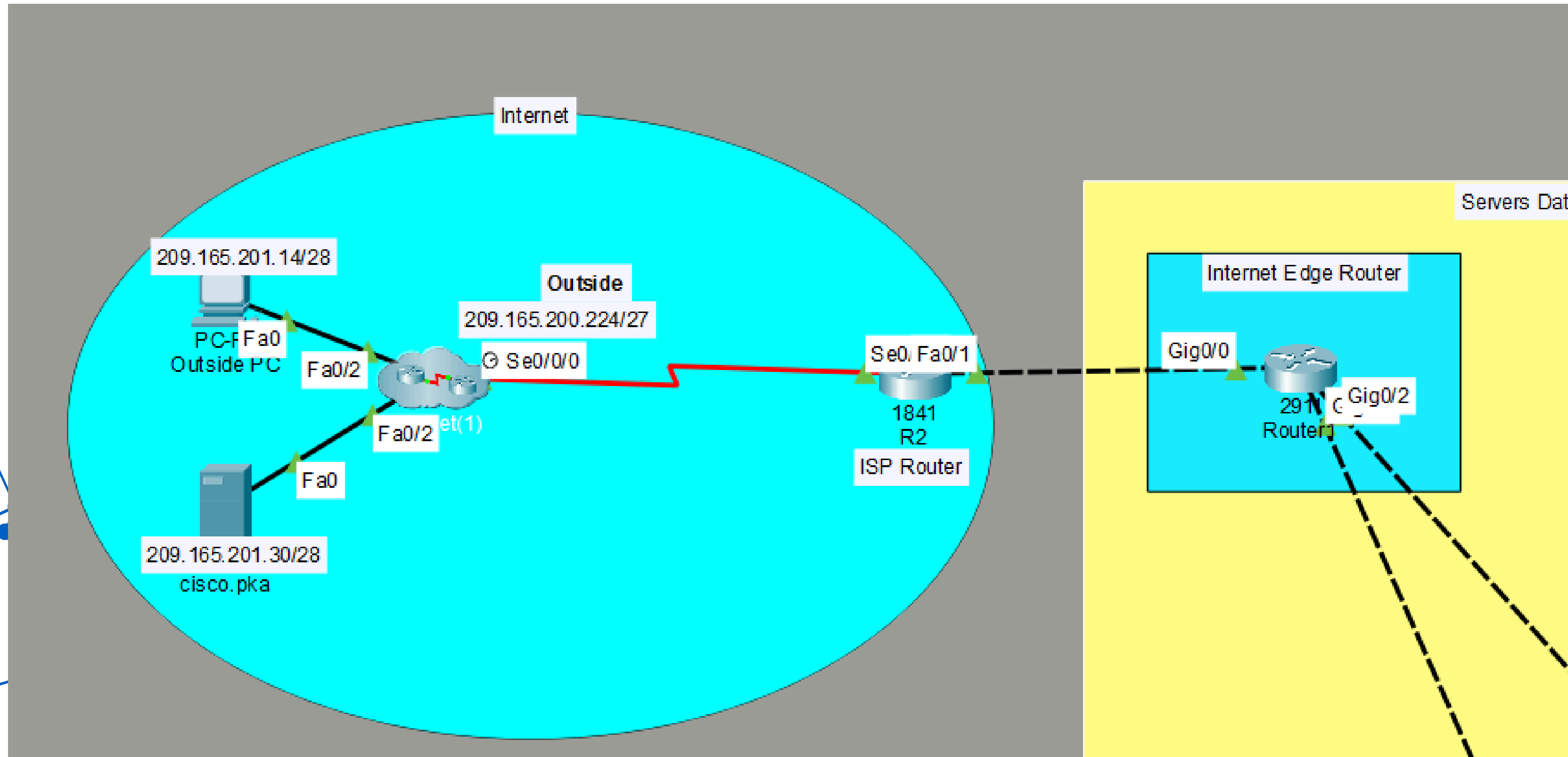
# WAN CORES BACKBONE OSPF



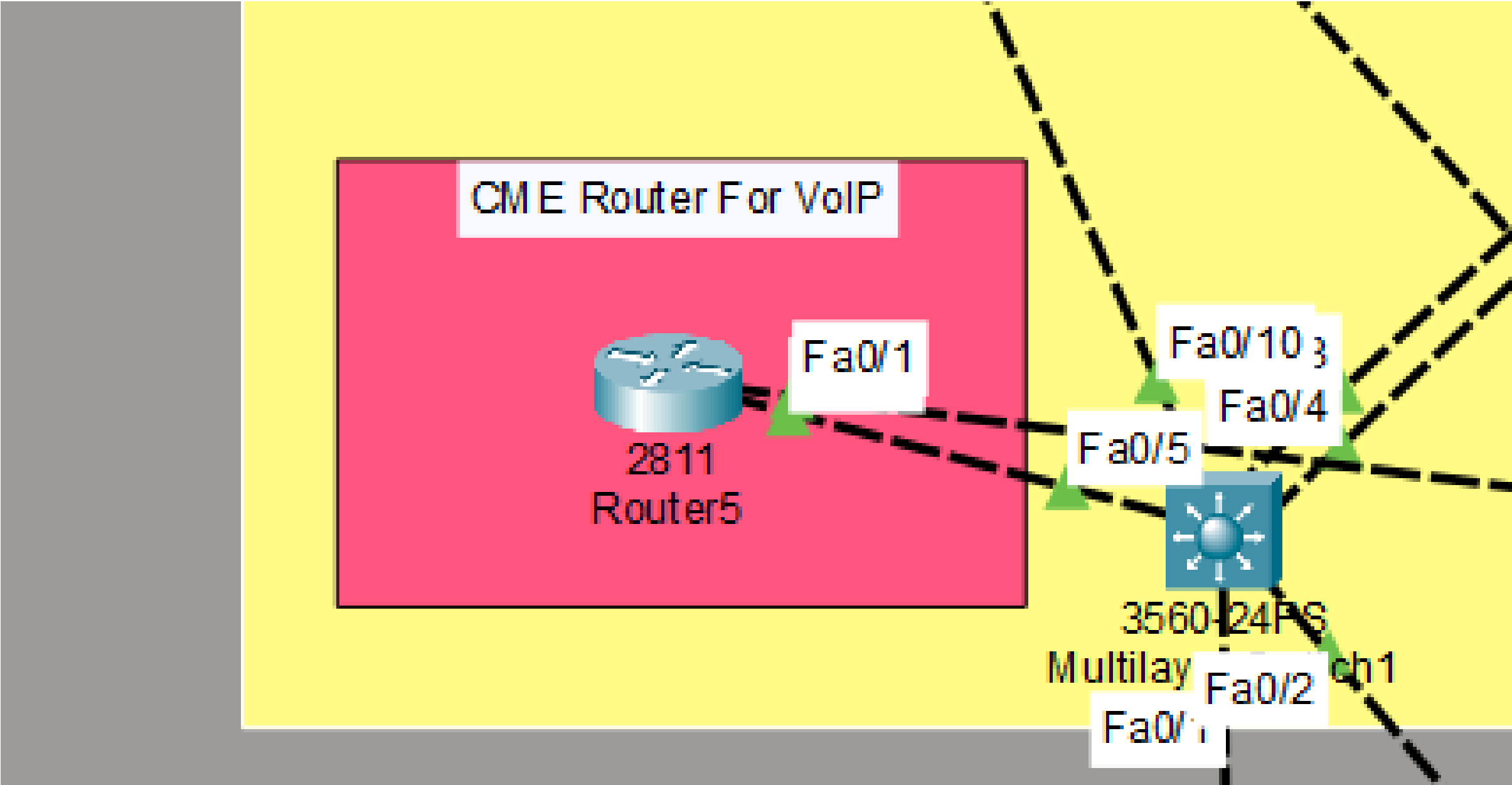
# ABRS IN EDGE ROUTER



# INTERNET EDGE ROUTER AND ISP

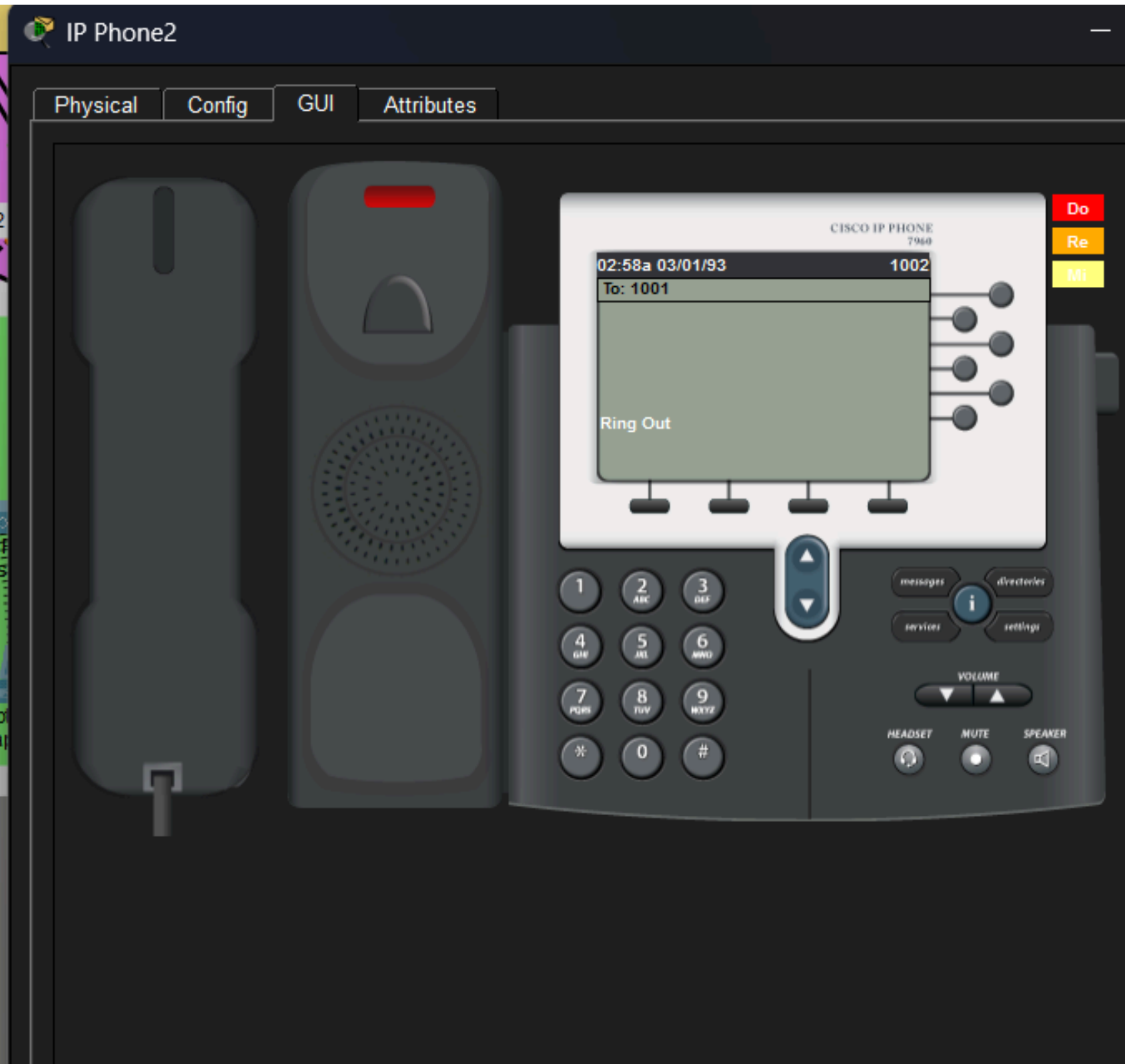
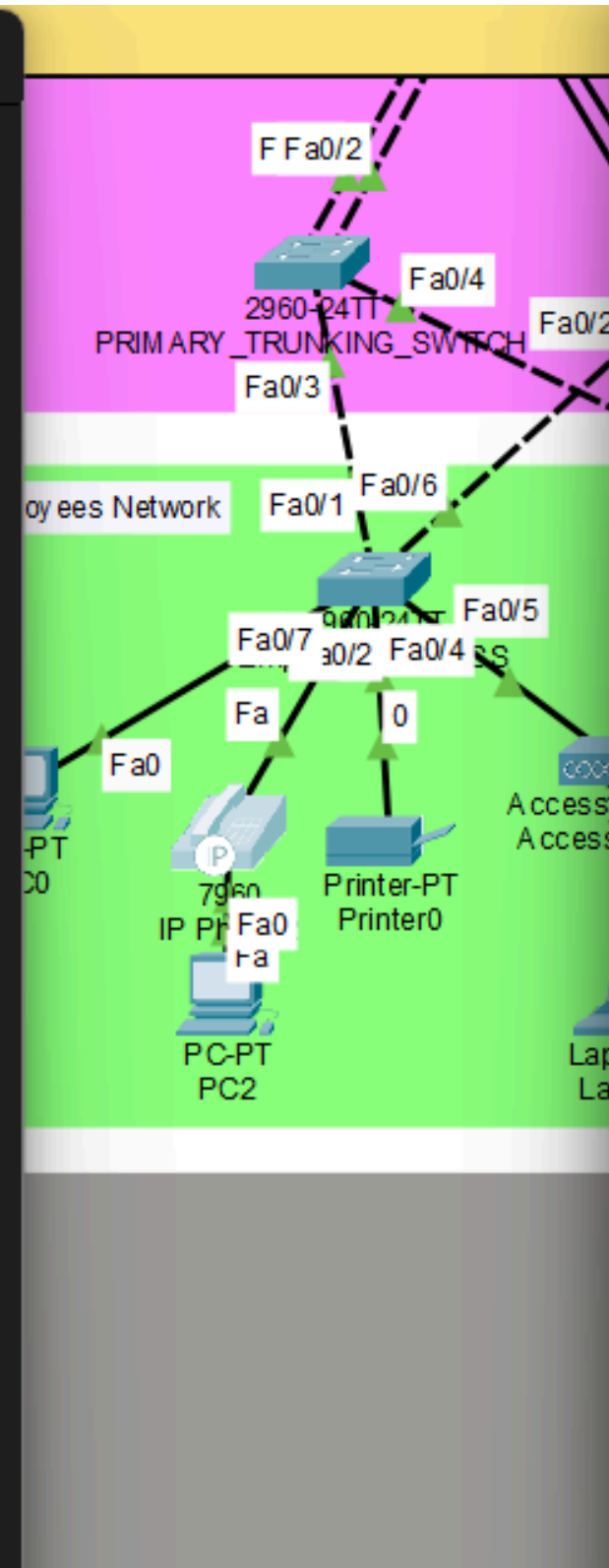
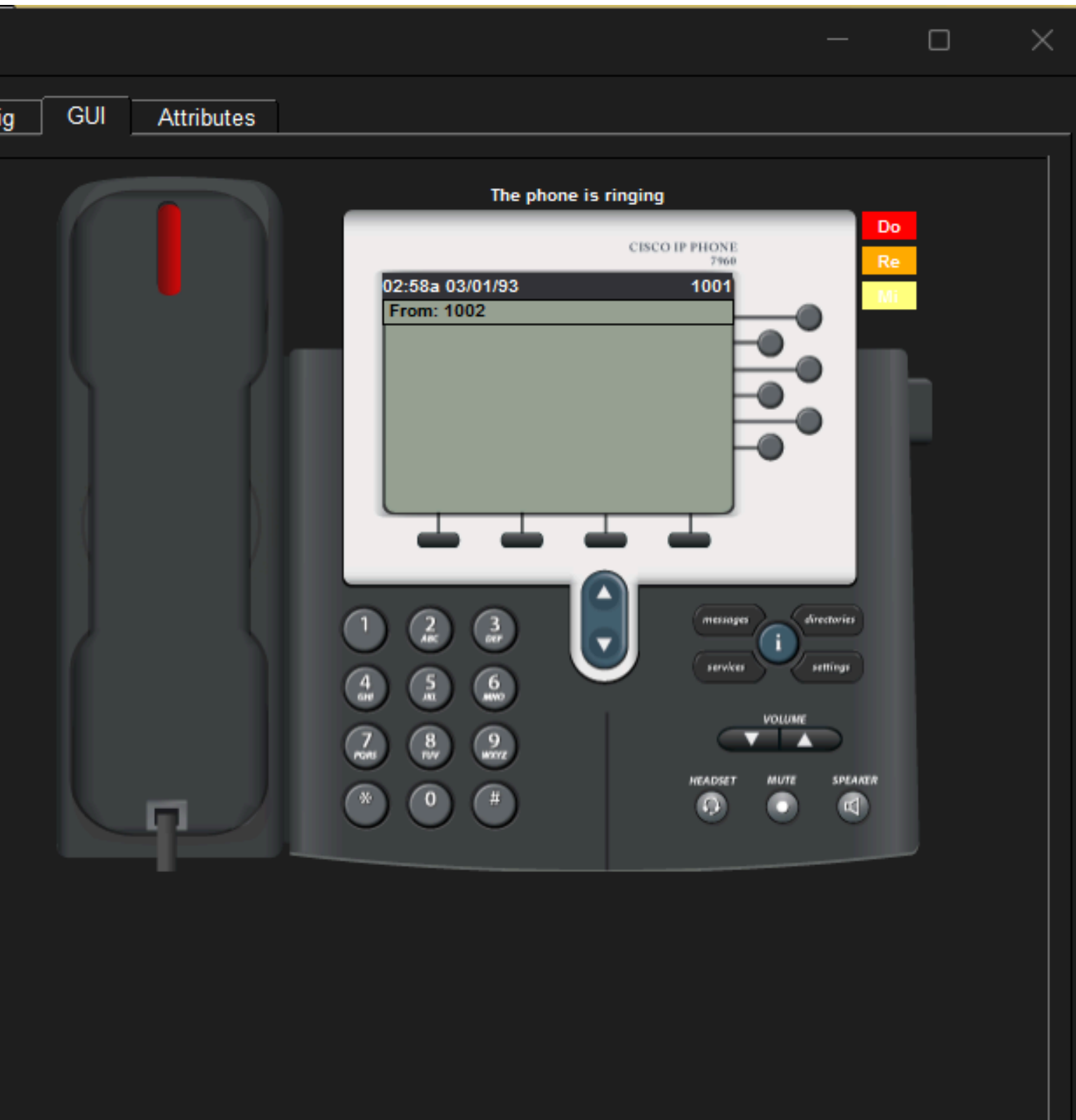


# CME ROUTER LOCATION



| Pool Name      | Default Gateway | DNS Server   | Start IP Address | Subnet Mask   | Max User | TFTP Server  | WLC Address |
|----------------|-----------------|--------------|------------------|---------------|----------|--------------|-------------|
| LUSAKA_VLAN_40 | 10.1.40.1       | 10.255.30.20 | 10.1.40.2        | 255.255.255.0 | 253      | 10.255.70.10 | 0.0.0.0     |

# TELEPHONES DIALING GUEST TO EMPLOYEE

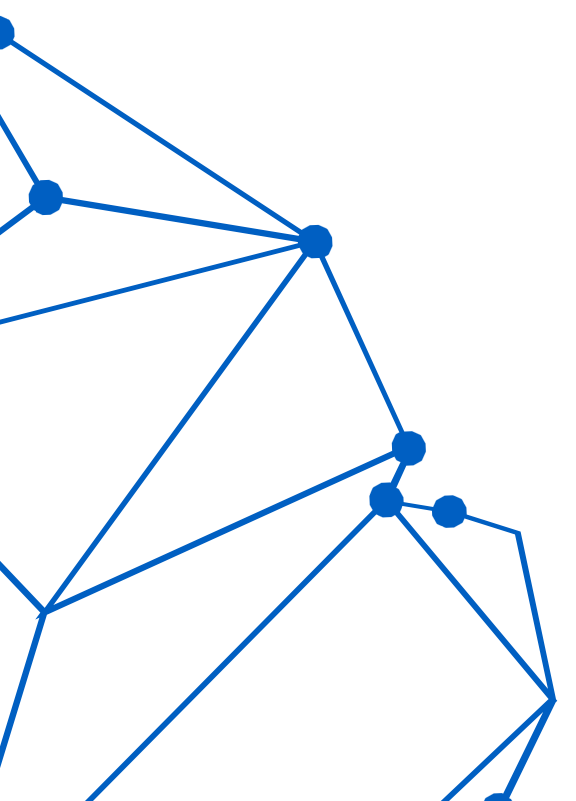


# DHCP SERVER CONFIGURATION



| Pool Name      | Default Gateway | DNS Server   | Start IP Address | Subnet Mask   | Max User | TFTP Server  | WLC Address |
|----------------|-----------------|--------------|------------------|---------------|----------|--------------|-------------|
| LUSAKA_VLAN_40 | 10.1.40.1       | 10.255.30.20 | 10.1.40.2        | 255.255.255.0 | 253      | 10.255.70.10 | 0.0.0.0     |
| LUSAKA_VLAN_99 | 10.1.99.1       | 10.255.30.20 | 10.1.99.2        | 255.255.255.0 | 253      | 0.0.0.0      | 0.0.0.0     |
| LUSAKA_VLAN_60 | 10.1.60.1       | 10.255.30.20 | 10.1.60.2        | 255.255.255.0 | 253      | 0.0.0.0      | 0.0.0.0     |
| LUSAKA_VLAN_50 | 10.1.50.1       | 10.255.30.20 | 10.1.50.2        | 255.255.255.0 | 253      | 0.0.0.0      | 0.0.0.0     |
| LUSAKA_VLAN_30 | 10.1.30.1       | 10.255.30.20 | 10.1.30.2        | 255.255.255.0 | 253      | 0.0.0.0      | 0.0.0.0     |
| LUSAKA_VLAN_10 | 10.1.8.1        | 10.255.30.20 | 10.1.8.2         | 255.255.248.0 | 2036     | 0.0.0.0      | 0.0.0.0     |
| LUSAKA_VLAN_20 | 10.1.16.1       | 10.255.30.20 | 10.1.16.2        | 255.255.252.0 | 1022     | 0.0.0.0      | 0.0.0.0     |

# DHCP SERVER TESTING



PC6

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 10.1.16.5

Subnet Mask 255.255.252.0

Default Gateway 10.1.16.1

DNS Server 10.255.30.20

# HTTP SERVER TRIAL



PC0

Physical Config Desktop Programming Attributes

Web Browser

URL <http://www.registry.zm> Go Stop

### Implemented Technologies

|                                  |                                               |                                         |                                      |
|----------------------------------|-----------------------------------------------|-----------------------------------------|--------------------------------------|
| <b>OSPF</b><br>Dynamic Routing   | <b>HSRP</b><br>Gateway Redundancy             | <b>EtherChannel</b><br>Link Aggregation | <b>VLANs</b><br>Network Segmentation |
| <b>ACLs</b><br>Security Policies | <b>DHCP &amp; DNS</b><br>Centralized Services | <b>VoIP</b><br>IP Telephony             | <b>AAA</b><br>Authentication         |

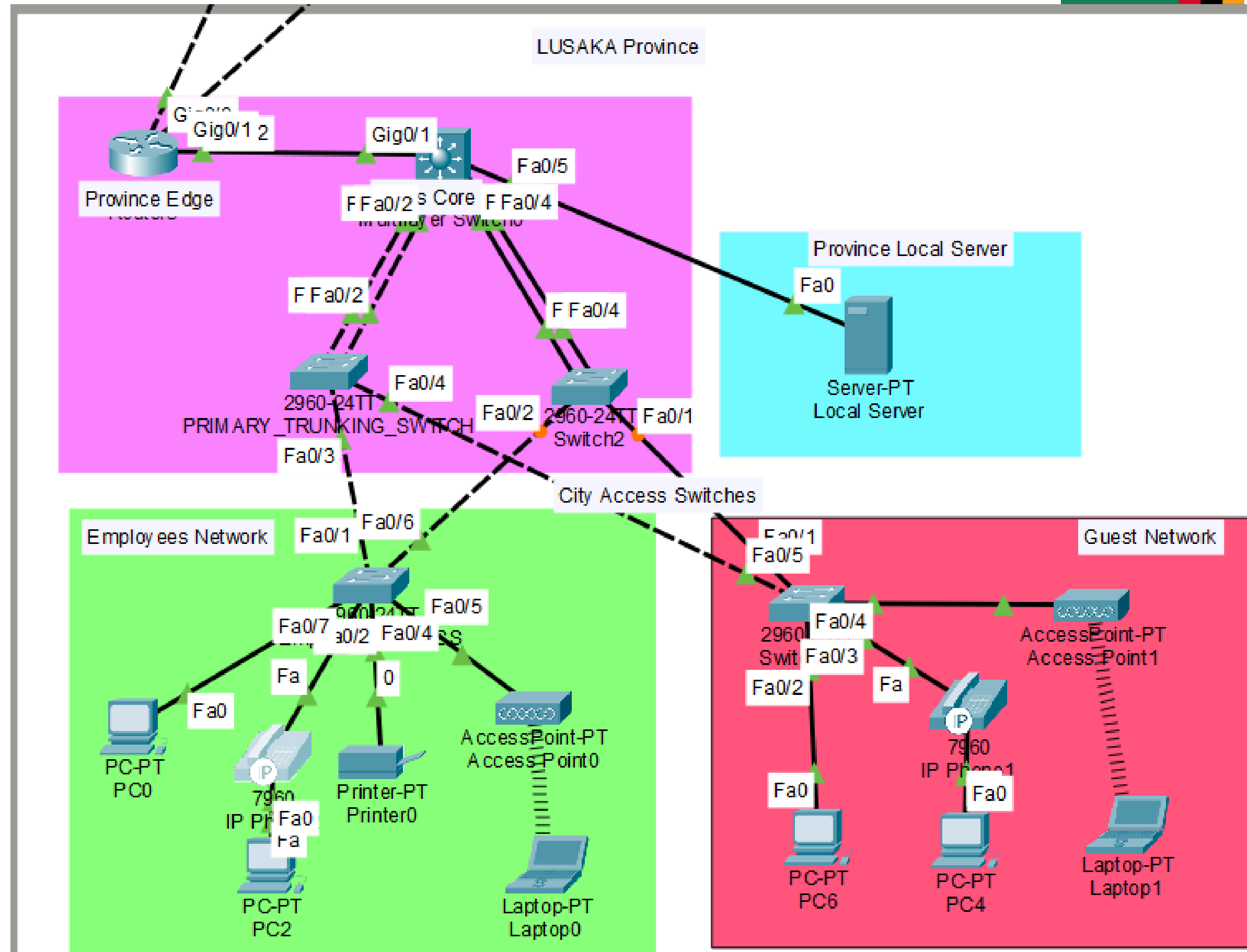
### Project Team

|                                                              |                                           |                                              |                                                    |
|--------------------------------------------------------------|-------------------------------------------|----------------------------------------------|----------------------------------------------------|
| <b>Mohamed Abdelhamid</b><br>Network Design & Implementation | <b>Ahmed Samir</b><br>Routing & Switching | <b>Mohamed Nael</b><br>Server Infrastructure | <b>Mohamed Nabil</b><br>Network Security & Testing |
|--------------------------------------------------------------|-------------------------------------------|----------------------------------------------|----------------------------------------------------|

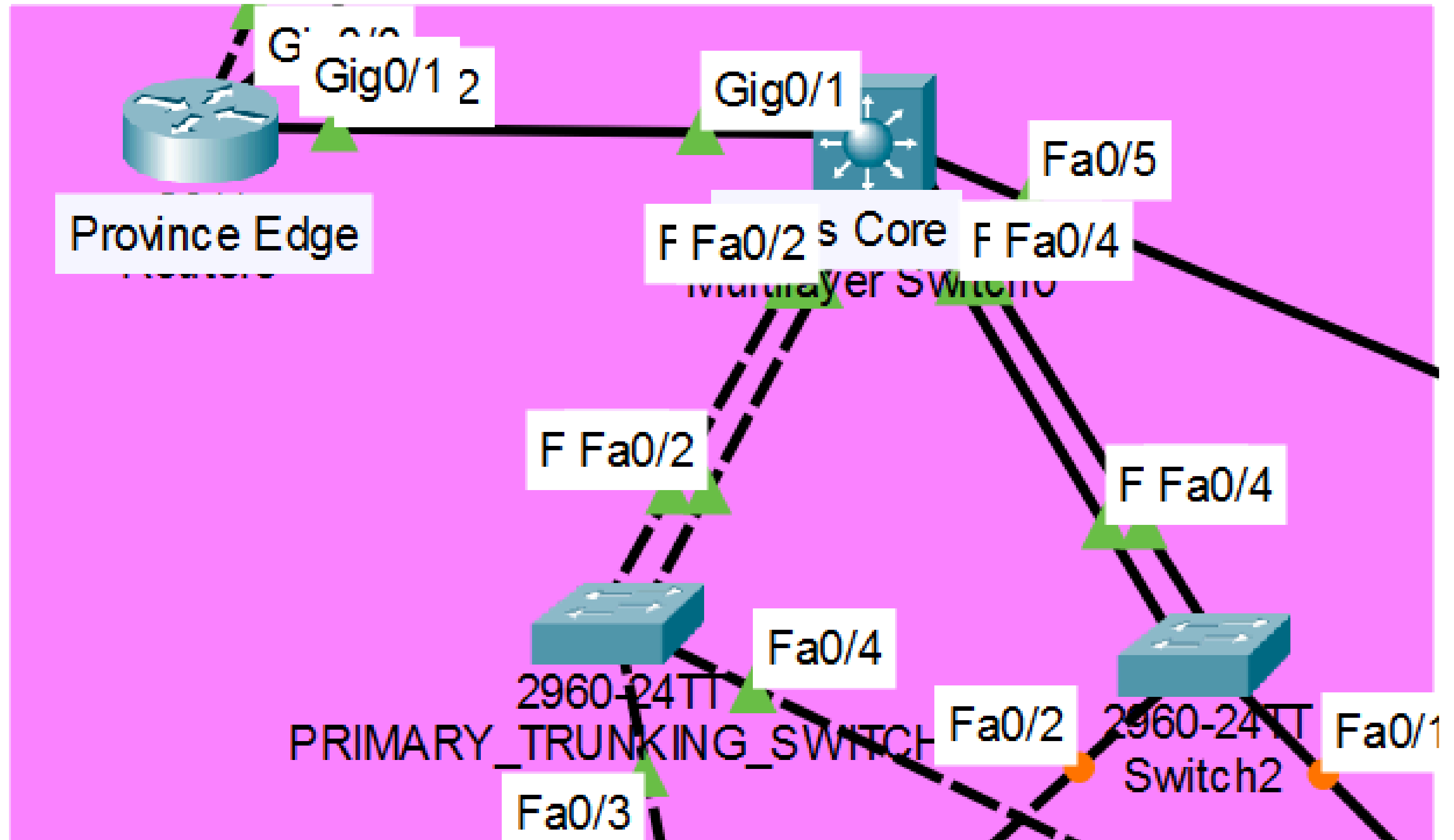
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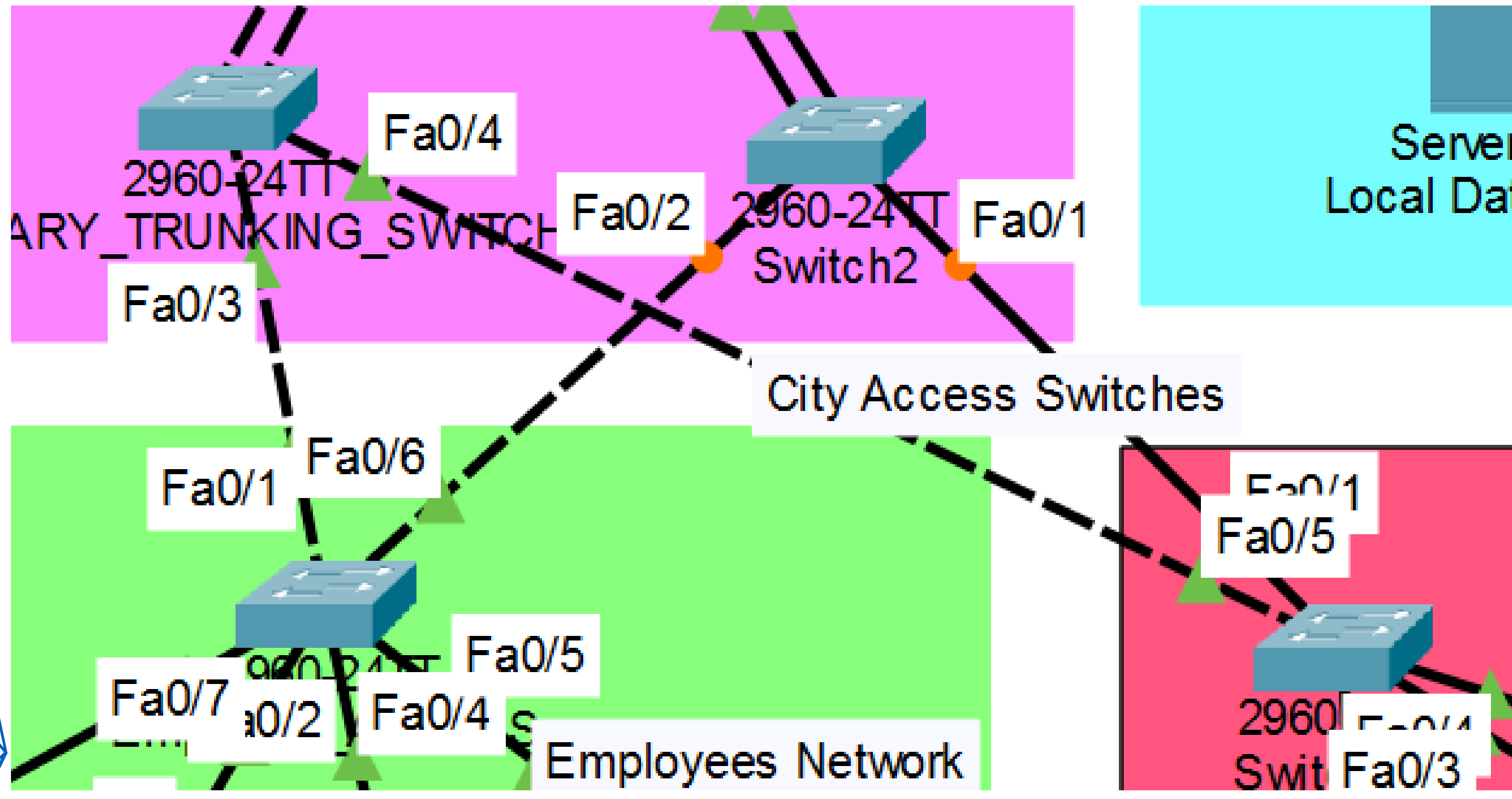
# PROVINCE ARCHITECTURE (LUSAKA)



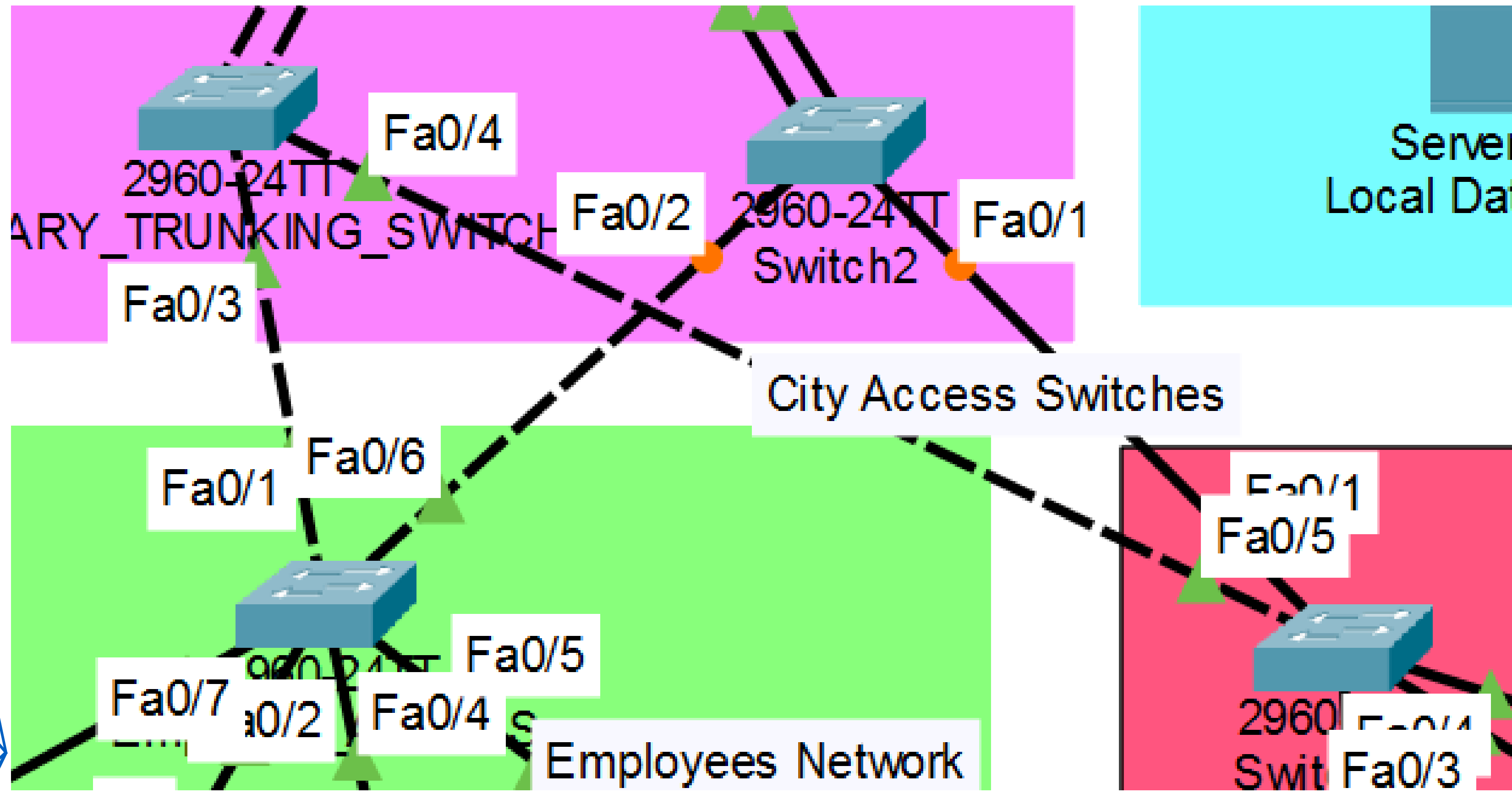
# ETHER-CHANNEL IN PRIM & SECONDARY



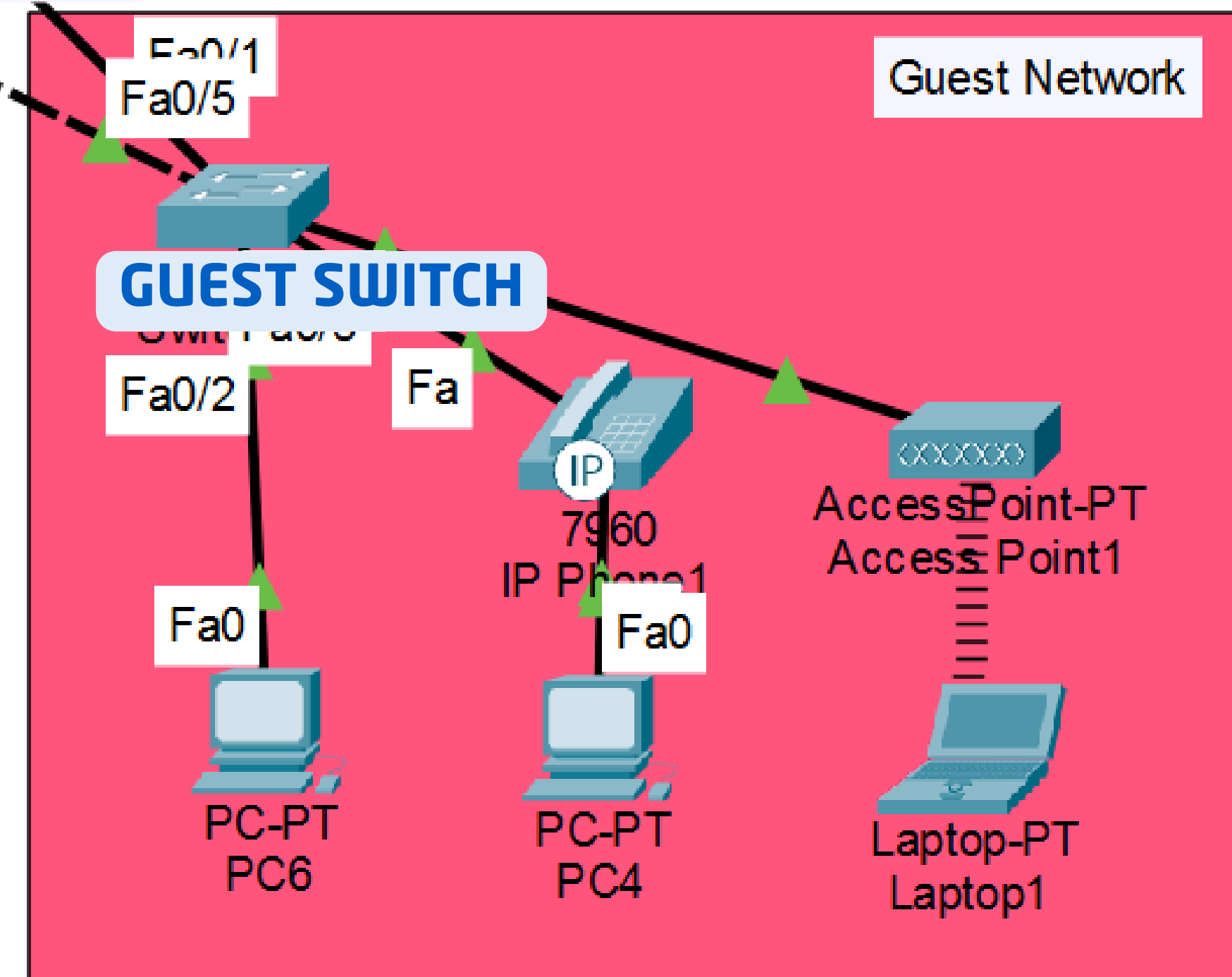
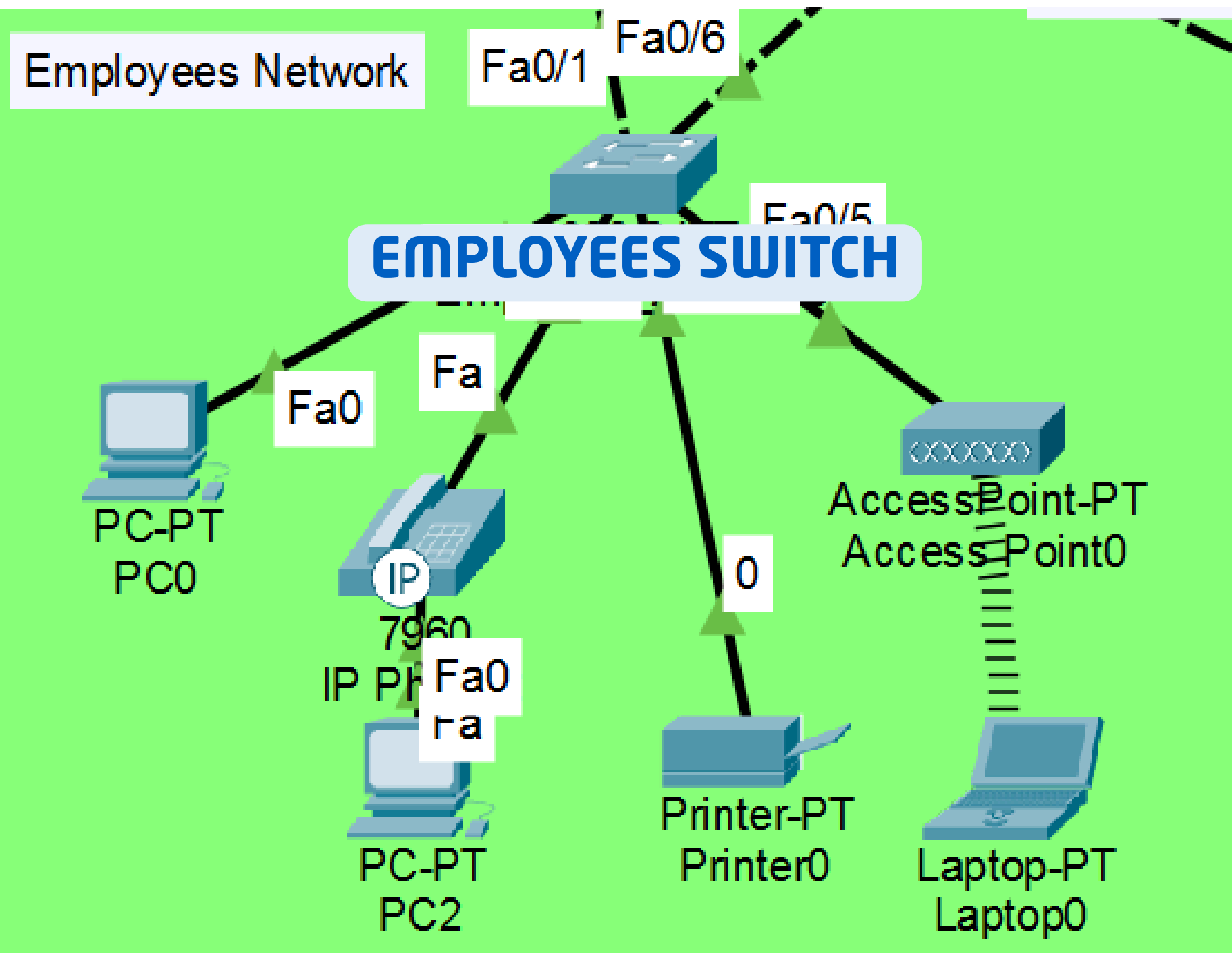
# PRIMARY & SECONDARY STP



# PRIMARY & SECONDARY STP



# ACCESS SWITCHES & DEVICES



# SECURITY IN ACCESS

**PORTSECURITY**

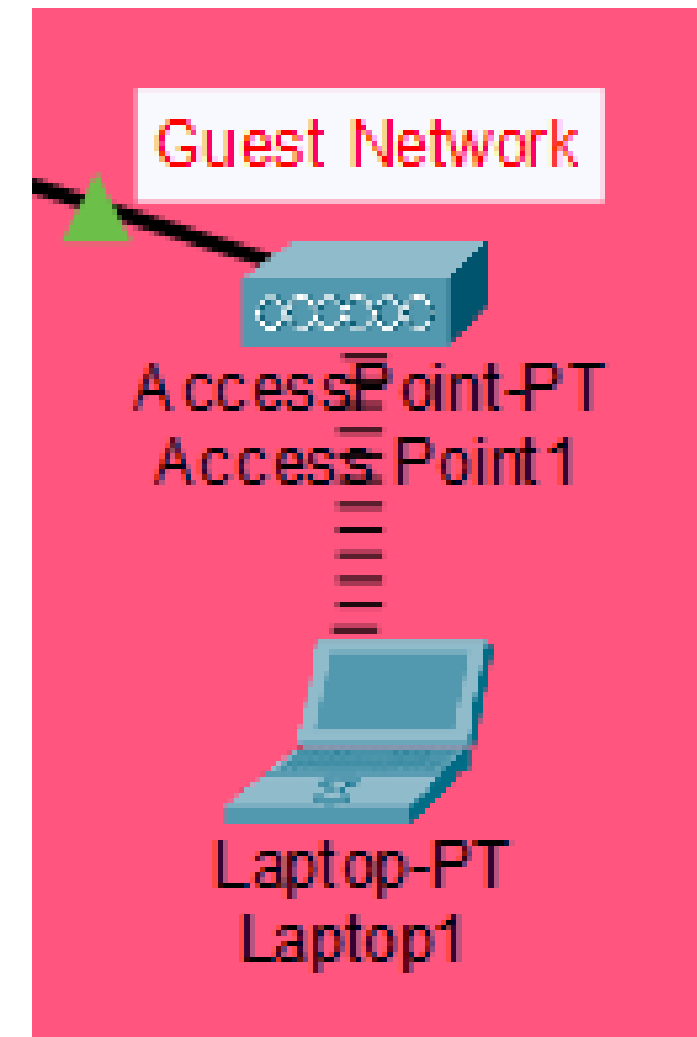
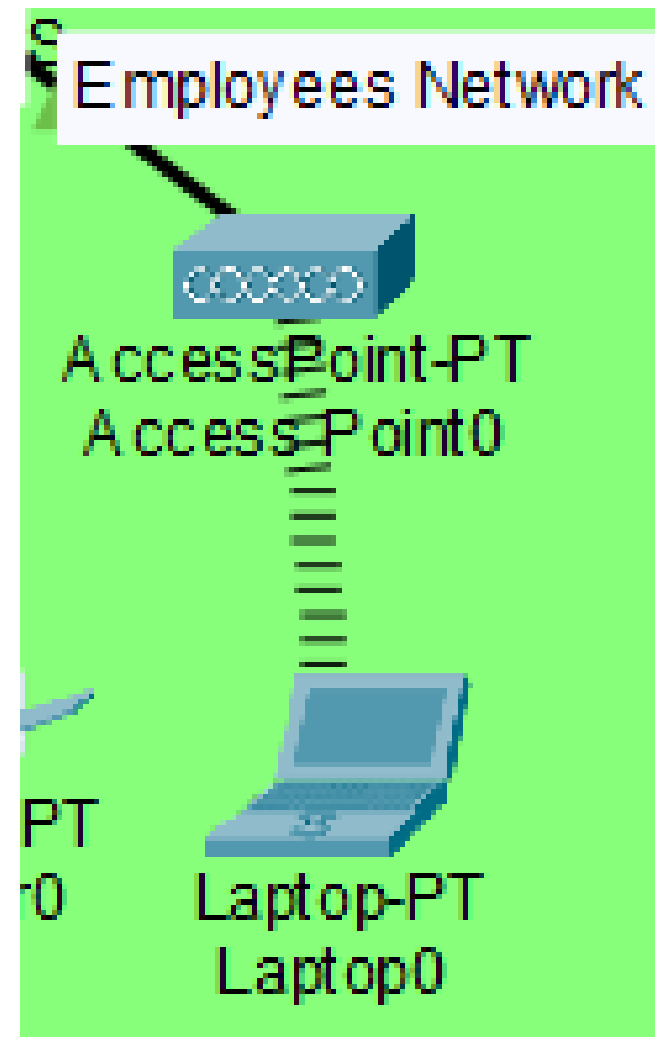
**BPDU GUARD**

**DAI DYNAMIC ARP INSPECTION**

**DHCP SPOOFING**



# WLAN DEVICES



**ALL PCS HERE IP CONFIGURED WITH DHCP**

# ACL



**WE USED EXTENDED ACL TO PREVENT GUEST FROM ANY OPERATIONS EXCEPT :**

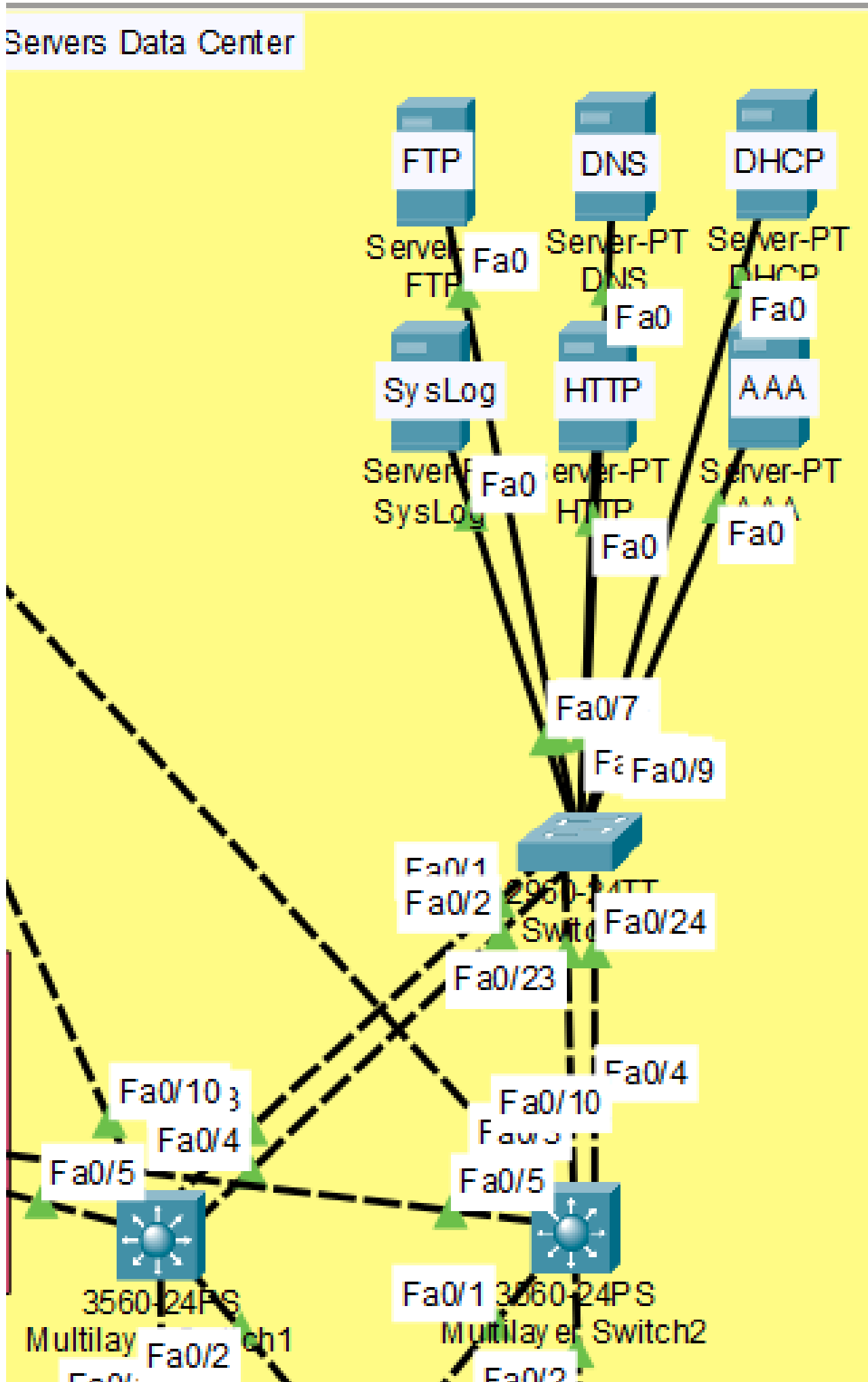
**1- DHCP FOR IP CONFIGURATION 10.255.30.10**

**2- DNS IP DOMAIN BOOK LIST**

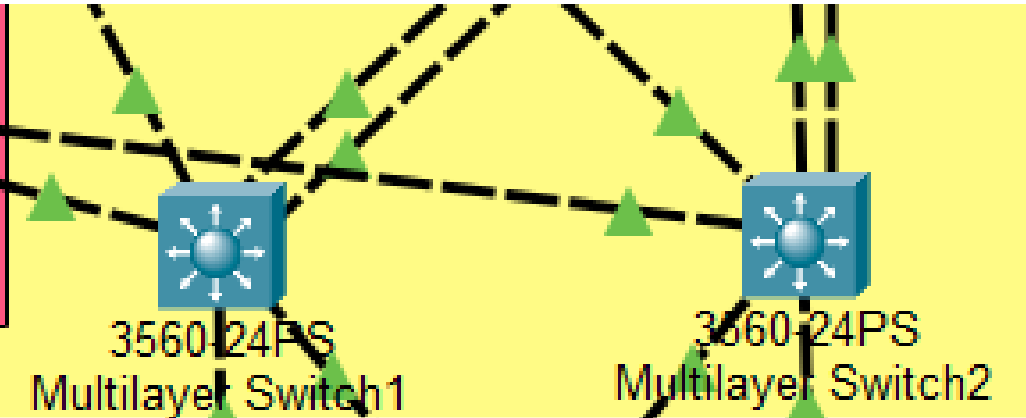
**3- HTTP FOR WEB BROWSING OPEN SOURCE CIVIL REGISTRY OPERATIONS**



# SERVICES TOPOLOGY



# HSRP IN SERVERS DATACENTER CORES



| Vlan | Vlan Gateway DCC1 | Vlan Gateway DCC2 | Vlan Virtual Gateway |
|------|-------------------|-------------------|----------------------|
| 10   | 10.255.8.2        | 10.255.8.3        | 10.255.8.1           |
| 20   | 10.255.16.2       | 10.255.16.3       | 10.255.16.1          |
| 30   | 10.255.30.2       | 10.255.30.3       | 10.255.30.1          |
| 40   | 10.255.40.2       | 10.255.40.3       | 10.255.40.1          |
| 50   | 10.255.50.2       | 10.255.50.3       | 10.255.50.1          |
| 60   | 10.255.60.2       | 10.255.60.3       | 10.255.60.1          |
| 99   | 10.255.99.2       | 10.255.99.3       | 10.255.99.1          |

# DNS SERVER CONFIGURATION



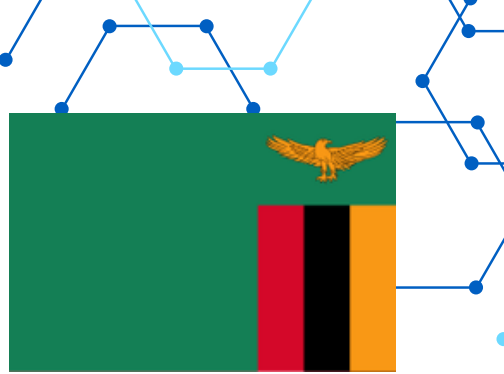
| Name             | Type     | Detail       |
|------------------|----------|--------------|
| aaa.registry.zm  | A Record | 10.255.30.40 |
| dhcp.registry.zm | A Record | 10.255.30.10 |
| ftp.registry.zm  | A Record | 10.255.30.30 |
| www.registry.zm  | A Record | 10.255.30.50 |

# FTP & DNS SERVERS TRIAL



```
C:\>ftp ftp.registry.zm
Trying to connect...ftp.registry.zm
Connected to ftp.registry.zm
220- Welcome to PT Ftp server
Username:admin
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>S|
```

# AAA SERVER CONFIGURATION & TESTING



|            | Client Name | Client IP   | Server Type | Key      |
|------------|-------------|-------------|-------------|----------|
| 1          | LUSAKA_CORE | 10.1.254.2  | Tacacs      | cisco123 |
| 2          | LUSAKA_EDGE | 10.254.1.2  | Tacacs      | cisco123 |
| 3          | WANCORE1    | 10.255.30.1 | Tacacs      | cisco123 |
|            |             |             |             |          |
| User Setup |             |             |             |          |
| Username   |             | Password    |             |          |
|            | Username    | Password    |             |          |
| 1          | admin       | admin       |             |          |

## LUSAKA EDGE ROUTER

User Access Verification

```
Username: admin
Password:
LUSAKA EDGE ROUTER>
```

## LUSAKA L3 CORE

User Access Verification

```
Username: admin
Password:
LUSAKA L3 SWITCH>
```

# SYSLOG SERVER LOGS SCREENSHOT



SysLog

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG**
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

Syslog

Service On Off

|     | Time | HostName    | Message                                                                  |
|-----|------|-------------|--------------------------------------------------------------------------|
|     |      |             | %HSRP-6-STATECHANGE: Vlan99 Grp 99 state Standby -> Active               |
| 138 | -    | 10.255.30.3 | 07:20:35: %OSPF-5-ADJCHG: Process 1, Nbr 10.255.70.10 on ...             |
| 139 | -    | 10.255.30.3 | 07:20:35: %OSPF-5-ADJCHG: Process 1, Nbr 10.255.70.10 on ...             |
| 140 | -    | 10.252.3.1  | 08:31:30: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.255.1 on ...               |
| 141 | -    | 10.252.3.1  | 08:31:30: OSPF: Build router LSA for area 0, router ID 10.255.0.1, se... |
| 142 | -    | 10.252.3.1  | 08:31:30: OSPF: DR/BDR election on FastEthernet0/1                       |
| 143 | -    | 10.252.3.1  | 08:31:30: OSPF: Elect DR 10.255.0.1                                      |
| 144 | -    | 10.252.3.1  | 08:31:30: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on ...                  |
| 145 | -    | 10.252.3.1  | 08:31:30: OSPF: Build router LSA for area 0, router ID 10.255.0.1, se... |
| 146 | -    | 10.252.3.1  | 08:31:30: OSPF: DR/BDR election on FastEthernet0/4                       |
| 147 | -    | 10.255.30.3 | %HSRP-6-STATECHANGE: Vlan30 Grp 30 state Speak -> Standby                |
| 148 | -    | 10.255.30.3 | %HSRP-6-STATECHANGE: Vlan99 Grp 99 state Speak -> Standby                |
| 149 | -    | 10.254.1.2  | %SYS-5-CONFIG_I: Configured from console by console                      |
| 150 | -    | 10.254.1.2  | %SYS-5-CONFIG_I: Configured from console by console                      |

Clear Log

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# SNMP SERVER

## SNMP CONFIGURATION

```
enable
configure terminal

snmp-server community CIVILREG-RO ro
snmp-server location Lusaka
snmp-server contact Network-Admin

end
write memory
```

## SHOW SNMP ON WAN CORE 1

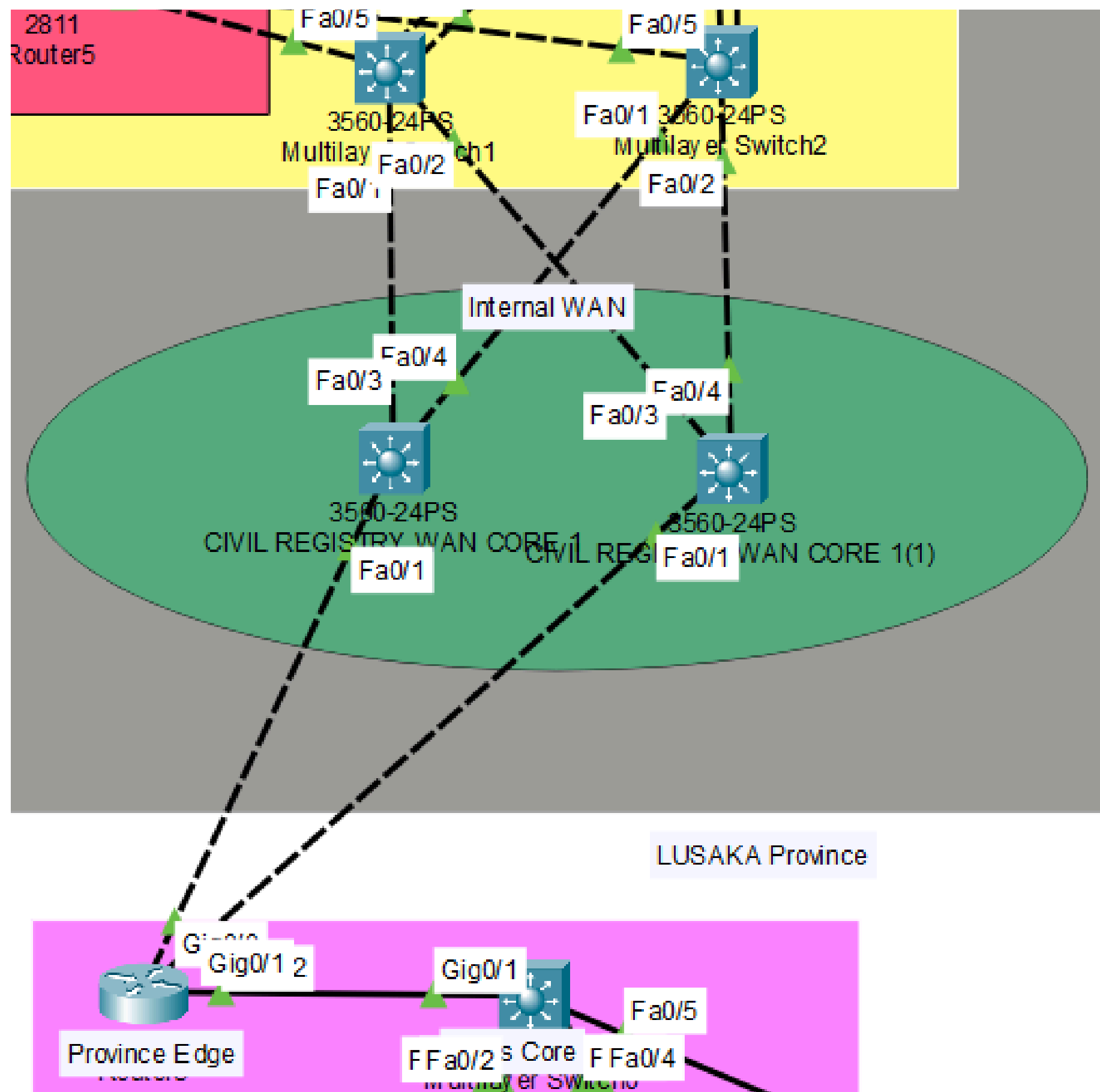
```
WAN_CORE_1(config)#do show running-config | include snmp
snmp-server community CIVILREG-RO RO
```



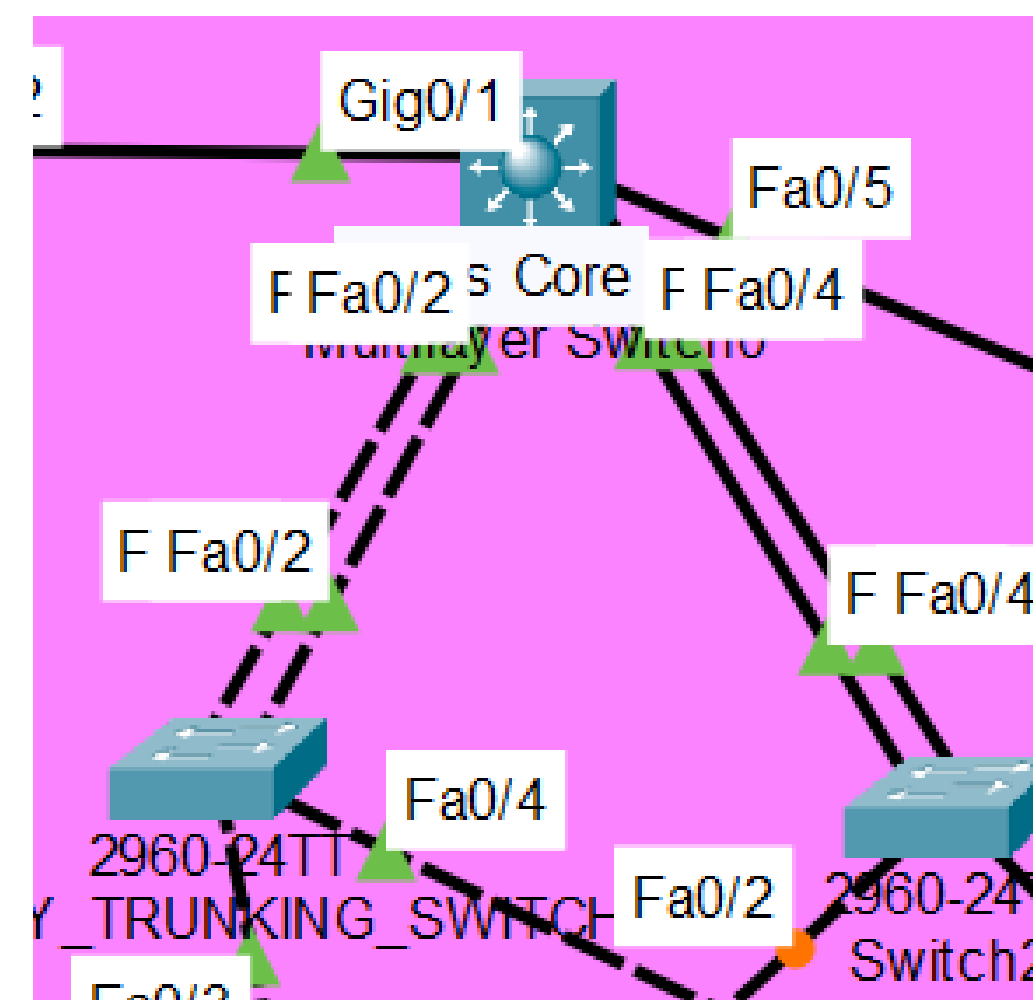
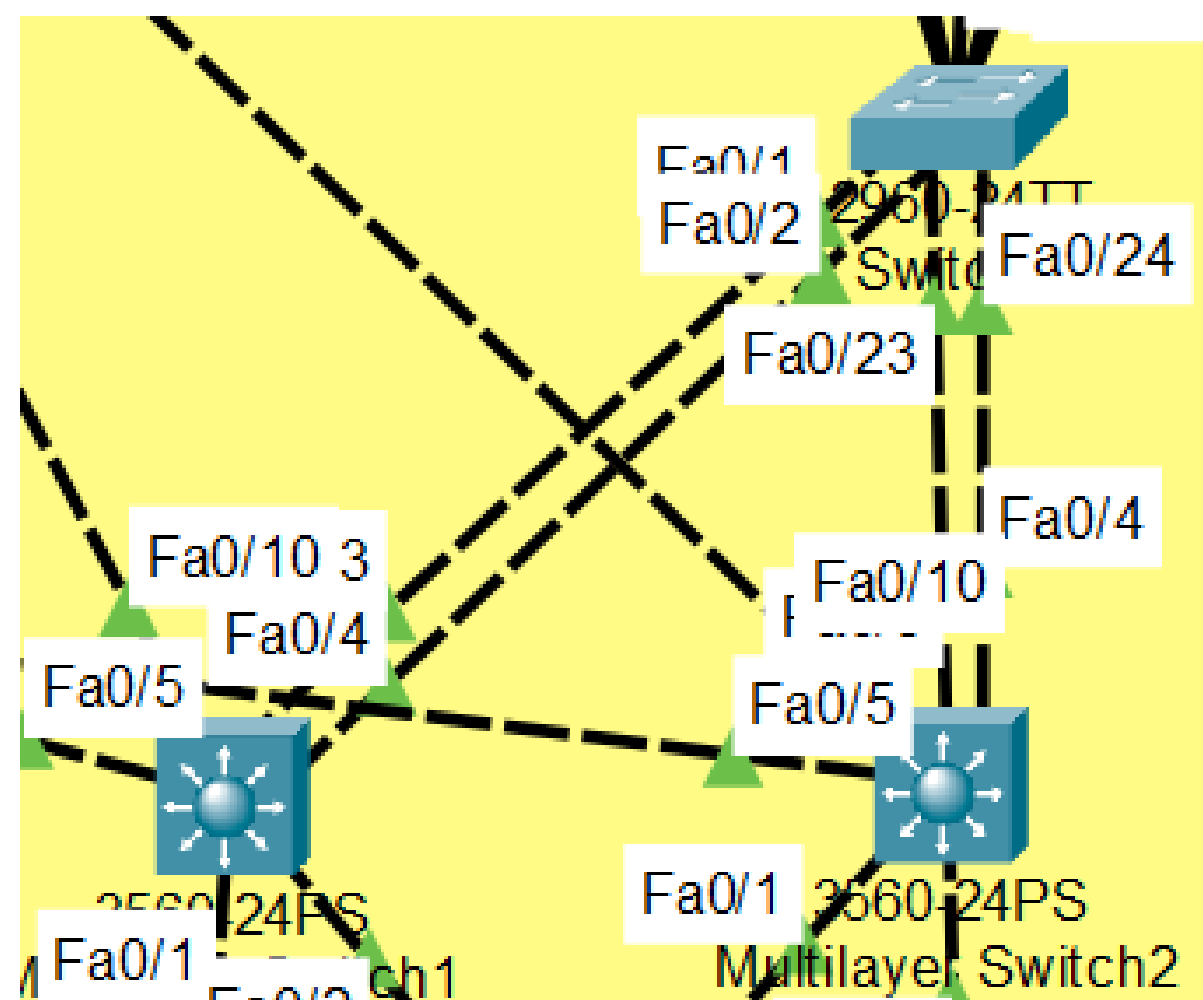
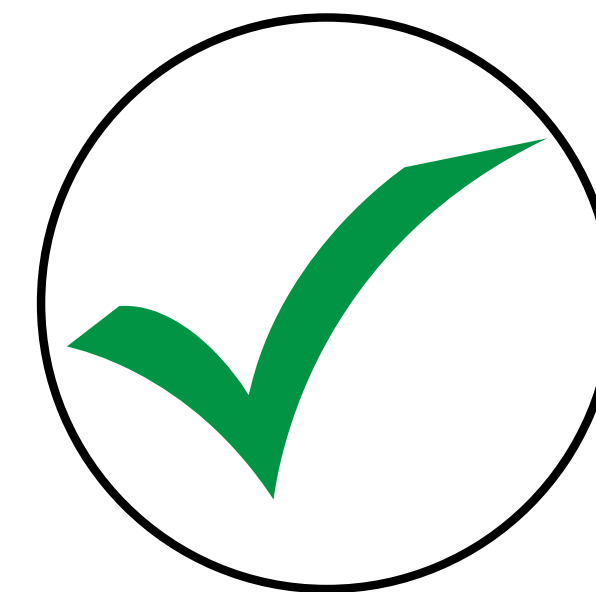
The image features a white background with decorative geometric patterns in the corners. The top-left and bottom-left corners contain blue line art of interconnected polygons, with small blue dots at the vertices. The top-right and bottom-right corners contain a honeycomb pattern of hexagons, with some hexagons outlined in a lighter blue color. Centered in the middle of the image is the main text.

# **ADVANTAGES & DISADVANTAGES & WHAT WE DIDNT INCLUDE**

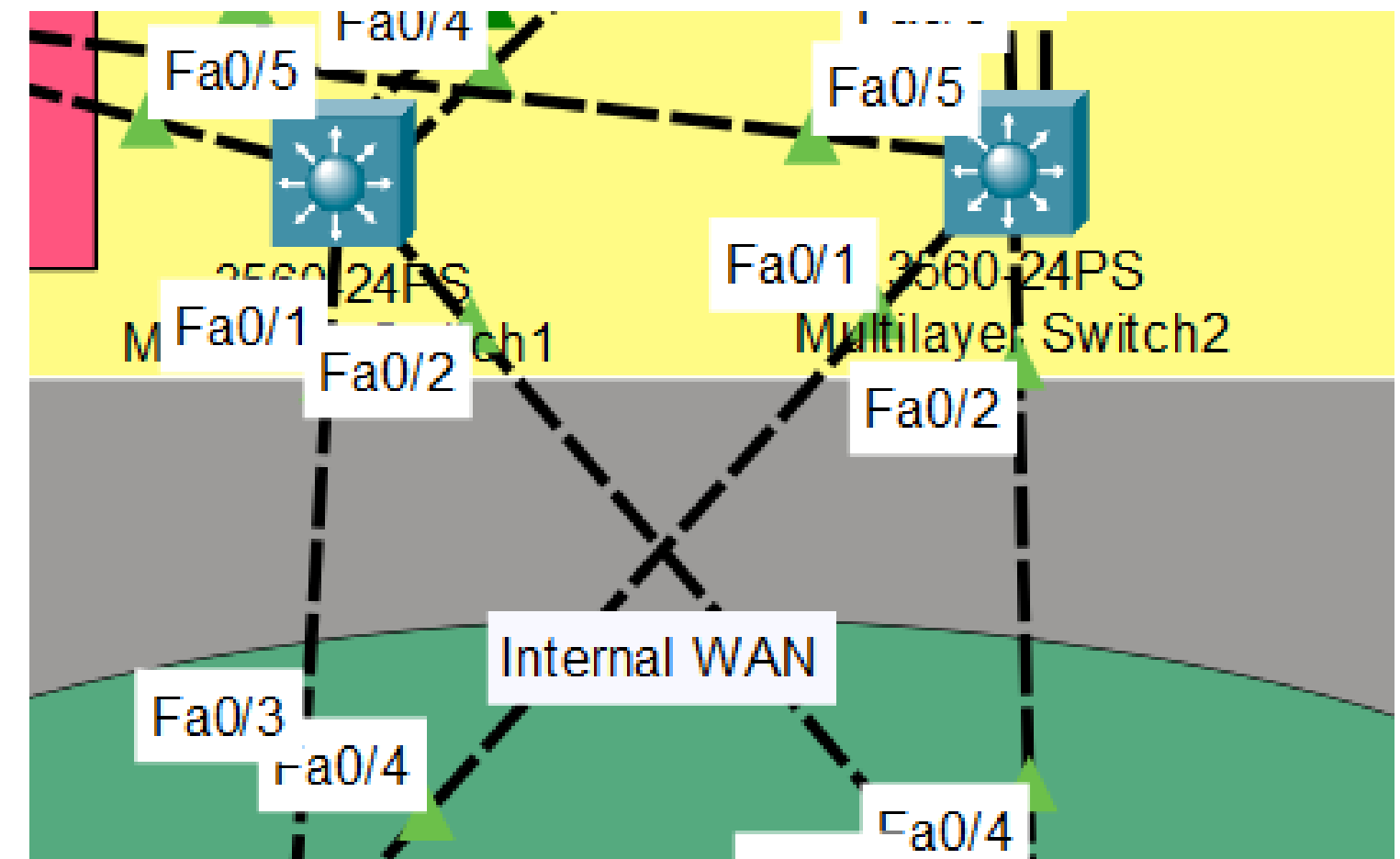
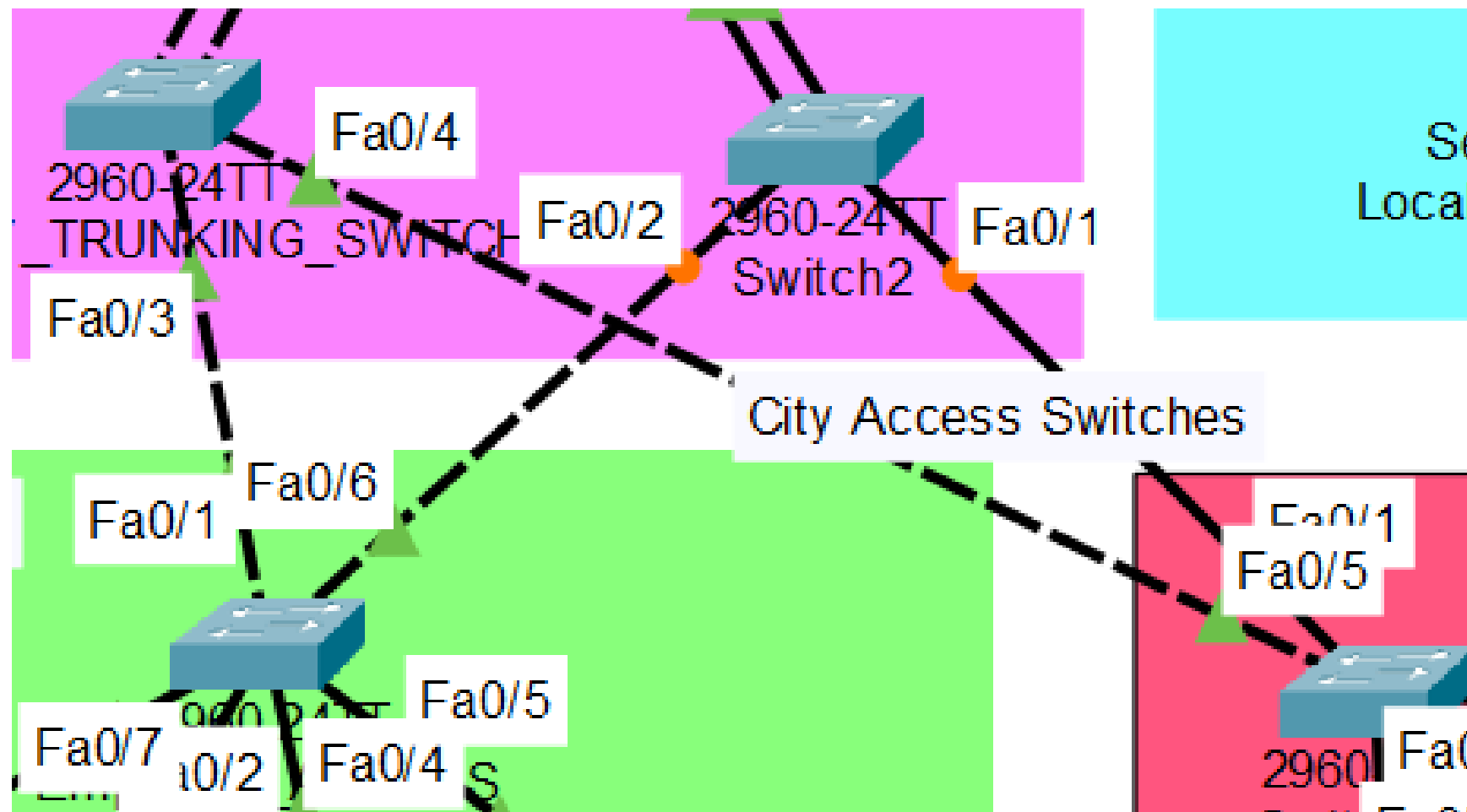
# ADVANTAGE BACKBONE



# ADVANTAGE MORE BANDWIDTH



# ADVANTAGE REDUNDANT



# ADVANTAGE CAPACITY PER PROVINCE SCALABLE PLANNING



*ip per province*



65634

# DISADVANTAGES

